

SCHOOL OF COMPUTER SCIENCE AND ARTIFICIAL INTELLIGENCE		DEPARTMENT OF COMPUTER SCIENCE ENGINEERING																		
Program Name: B. Tech		Assignment Type: Lab	Academic Year:2025-2026																	
Course Coordinator Name		Dr. Rishabh Mittal																		
Instructor(s) Name		<table border="1"> <tr><td>Mr. S Naresh Kumar</td></tr> <tr><td>Ms. B. Swathi</td></tr> <tr><td>Dr. Sasanko Shekhar Gantayat</td></tr> <tr><td>Mr. Md Sallauddin</td></tr> <tr><td>Dr. Mathivanan</td></tr> <tr><td>Mr. Y Srikanth</td></tr> <tr><td>Ms. N Shilpa</td></tr> <tr><td>Dr. Rishabh Mittal (Coordinator)</td></tr> <tr><td>Dr. R. Prashant Kumar</td></tr> <tr><td>Mr. Ankushavali MD</td></tr> <tr><td>Mr. B Viswanath</td></tr> <tr><td>Ms. Sujitha Reddy</td></tr> <tr><td>Ms. A. Anitha</td></tr> <tr><td>Ms. M.Madhuri</td></tr> <tr><td>Ms. Katherashala Swetha</td></tr> <tr><td>Ms. Velpula sumalatha</td></tr> <tr><td>Mr. Bingi Raju</td></tr> </table>		Mr. S Naresh Kumar	Ms. B. Swathi	Dr. Sasanko Shekhar Gantayat	Mr. Md Sallauddin	Dr. Mathivanan	Mr. Y Srikanth	Ms. N Shilpa	Dr. Rishabh Mittal (Coordinator)	Dr. R. Prashant Kumar	Mr. Ankushavali MD	Mr. B Viswanath	Ms. Sujitha Reddy	Ms. A. Anitha	Ms. M.Madhuri	Ms. Katherashala Swetha	Ms. Velpula sumalatha	Mr. Bingi Raju
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CourseCode	23CS002PC304	Course Title	AI Assisted Coding																	
Year/Sem	III/II	Regulation	R23																	
Date and Day of Assignment	Week9 – Monday	BATCH	B-48																	
NAME	M.Anu Chandra	H.T.NO	2303A54070																	
Assignment Number: 18.1(Present assignment number)/24(Total number of assignments)																				
Q.No.	Question	Expected Time to complete																		
1	Lab 18 – API Integration: Connecting to External Services with Error Handling	Week9 - Monday																		

	Lab Objectives: • Learn how to integrate Python programs with external REST APIs. • Understand API request/response handling using requests or similar libraries. • Implement proper error handling for failed API calls (timeouts, invalid responses, rate limits). • Practice extracting and displaying meaningful information from API responses.	
	Task 1 – Weather API Integration Connect to a Weather API to fetch current weather details for a given city. Instructions: • Prompt AI to generate Python code using the requests library. • Handle errors: invalid city name, API key errors, and network failures. • Display temperature, humidity, and weather description in a readable format. PROMPT : Explain how to integrate a Weather API in Python to fetch real-time weather data for a given city. Describe how API requests are sent, responses are processed, and how error handling is implemented for invalid inputs and network failures.	

```
1 import requests
2
3 # =====
4 # CONFIGURATION
5 # =====
6 API_KEY = "YOUR_API_KEY_HERE" # Replace with your API key
7 BASE_URL = "https://api.openweathermap.org/data/2.5/weather"
8
9
10 def get_weather(city):
11     """Fetch weather data for a given city"""
12
13     params = {
14         "q": city,
15         "appid": API_KEY,
16         "units": "metric" # Celsius
17     }
18
19     try:
20         # Send request
21         response = requests.get(BASE_URL, params=params, timeout=10)
22
23         # Check HTTP errors
24         response.raise_for_status()
25
26         data = response.json()
27
28         # Handle API errors
29         if data.get("cod") != 200:
30             return f"Error: {data.get('message', 'Unknown error')}"
31
32     except requests.exceptions.RequestException as e:
33         return f"Request failed: {e}"
34
35
36 # Example usage
37 city = input("Enter city name: ")
38 weather = get_weather(city)
39 print(weather)
```

Expected Output:

- A working Python script that fetches and displays weather data with proper error handling.

EXPLANATION : In this task, we connect a Python program to an external Weather API using the `requests` library. The API sends back data in JSON format when we provide a city name and API key. The program extracts useful details such as temperature, humidity, and weather description from the response. Error handling is very important here:

- If the city name is incorrect → API returns an error.
- If API key is invalid → authentication error occurs.
- If network fails → request exception is handled.

This ensures the program runs smoothly without crashing and gives meaningful messages to the user.

Task 2 – Currency Exchange Rates

Integrate a Currency Exchange API to convert INR to USD, EUR, and GBP.

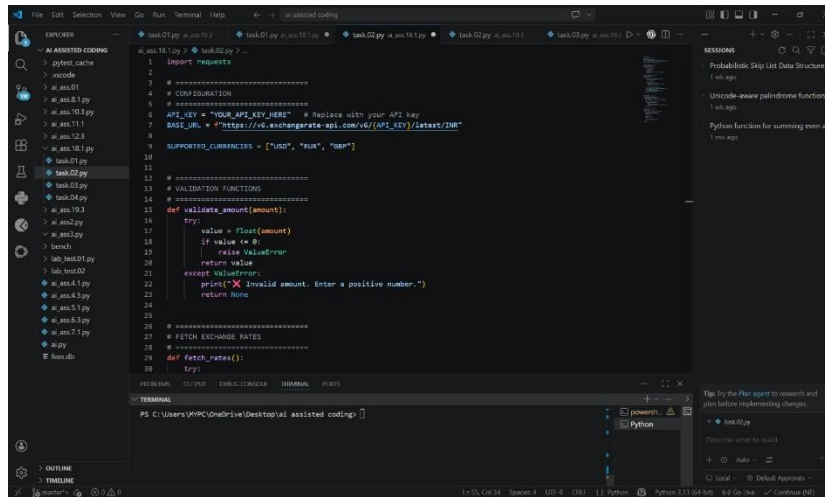
Instructions:

- Use AI to generate API request code.
- Validate user input for amount and currency codes.
- Implement error handling for invalid responses and API

downtime.

- Show conversion results clearly in tabular format.

PROMPT : Describe how to use a Currency Exchange API to convert INR into multiple currencies. Explain input validation, API request handling, and error management for incorrect inputs and API failures.



```
1 import requests
2
3 # =====
4 # CONFIGURATION
5 # =====
6 API_KEY = "YOUR_API_KEY_HERE" # Replace with your API key
7 BASE_URL = "https://v6.exchangerate-api.com/v6/{API_KEY}/latest/INR"
8
9 SUPPORTED_CURRENCIES = ["USD", "EUR", "GBP"]
10
11 # =====
12 # VALIDATION FUNCTIONS
13 # =====
14 def validate_amount(amount):
15     try:
16         value = float(amount)
17         if value <= 0:
18             raise ValueError
19         return value
20     except ValueError:
21         print("Invalid amount. Enter a positive number.")
22         return None
23
24 # =====
25 # RETURN EXCHANGE RATES
26 # =====
27 def fetch_rates():
28     try:
29         response = requests.get(BASE_URL.format(API_KEY))
30         data = response.json()
31         rates = data['rates']
32         return rates
33     except requests.exceptions.RequestException as e:
34         print(f"API request failed: {e}")
35         return None
36
37 # =====
38 # MAIN LOGIC
39 # =====
40 def main():
41     print("Currency Converter")
42     print("Enter amount in INR: ")
43     amount = input()
44     validated_amount = validate_amount(amount)
45     if validated_amount is None:
46         return
47     rates = fetch_rates()
48     if rates is None:
49         print("API request failed. Please try again later.")
50         return
51     print("\nExchange Rates:")
52     for currency, rate in rates.items():
53         print(f"{currency}: {rate}")
54
55 if __name__ == "__main__":
56     main()
```

Expected Output:

- A script that converts INR to multiple currencies with graceful error handling.

EXPLANATION : This task focuses on converting Indian Rupees (INR) into currencies like USD, EUR, and GBP using an external API.

Steps involved:

- Take user input (amount in INR)
- Validate input (must be numeric and positive)
- Send API request to get exchange rates
- Perform conversion using received rates

Error handling includes:

- Invalid input (non-numeric values)
- API downtime or failure
- Incorrect currency codes

The result is displayed clearly in a structured format (like a table), making it easy to understand.

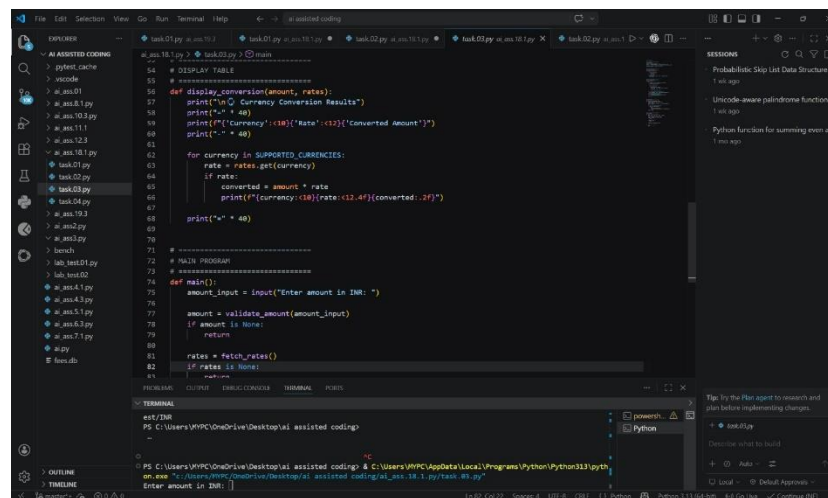
Task 3 – GitHub Repository Info Fetcher

Connect to the GitHub API to fetch details of a repository (e.g., stars, forks, issues).

Instructions:

- Prompt AI to write a Python function to send GET requests to GitHub API.
- Handle rate limit errors, 404 (repo not found), and invalid input.
- Display repository name, description, stars, forks, and open issues.

PROMPT : Explain how to fetch repository details from GitHub API using Python. Describe how GET requests work and how errors like rate limits, invalid repositories, and missing data are handled.



```
56 # DISPLAY TABLE
57 # =====
58 def display_conversion(amount, rates):
59     print("\n Currency Conversion Results")
60     print("-" * 40)
61     print(f"Currency '{rates['rate']}' (Rate: {rates['converted_amount']})")
62     print("-" * 40)
63
64 for currency in SUPPORTED_CURRENCIES:
65     rate = rates.get(currency)
66     if rate:
67         converted = amount * rate
68         print(f"Currency: {rate} (Rate: {rate}, converted: {converted:.2f})")
69     print("-" * 40)
70
71 # =====
72 # MAIN PROGRAM
73 # =====
74 def main():
75     amount_input = input("Enter amount in INR: ")
76
77     amount = validate_amount(amount_input)
78     if amount is None:
79         return
80
81     rates = fetch_rates()
82     if rates is None:
83         return
84
85     display_conversion(amount, rates)
```

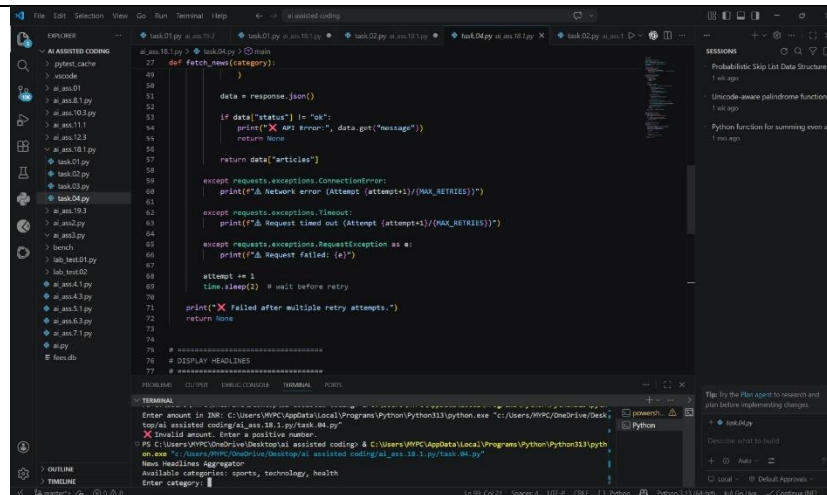
Expected Output:

- A Python program that retrieves and displays GitHub repository information with robust error handling.

EXPLANATION : In this task, we use the GitHub API to retrieve repository information such as:

- Repository name
- Description
- Stars

	• Forks • Open issues The Python program sends a GET request to the GitHub API endpoint. The response is in JSON format, from which required details are extracted. Error handling includes: • 404 Error → Repository not found • Rate limit exceeded → API blocks further requests • Invalid input → wrong repository name This makes the program reliable and user-friendly.	
	Task 4 – Real-Time Application: News Headlines Aggregator Scenario: Build a News Aggregator using a free News API. Requirements: • Fetch top 5 headlines for a given category (sports, technology, health). • Handle errors like invalid category, missing API key, and request failures. • Display headlines in a user-friendly numbered list format. • Include a retry mechanism if the first request fails. PROMPT : Explain how to build a news aggregator application using a News API. Discuss fetching headlines, handling errors, and implementing retry mechanisms for failed requests.	



Expected Output:

- A working Python script that retrieves and displays news headlines with proper error handling

EXPLANATION : This task involves creating a real-time application that fetches top news headlines based on categories like sports, technology, or health.

Steps:

- User selects a category
- API request is sent to fetch news
- Top 5 headlines are extracted and displayed

Error handling includes:

- Invalid category input
- Missing or wrong API key
- Network or request failure

A retry mechanism is implemented so that if the request fails initially, the program tries again automatically. This improves reliability.

The output is displayed in a numbered list, making it easy to read.